



Bridging the Protocol Gap: Real-Time Modbus-DNP3 Translation for Scalable DER Integration

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RESEARCH APPROACH

- ❖ **RAPID DER GROWTH & PROTOCOL MANDATES:** IEEE 1547-2018 requires DERs to use either of three standards SunSpec Modbus[1][2][3], DNP3 (IEEE 1815)[4] or IEEE 2030.5, yet deployments now mix all three
- ❖ **MIXED-PROTOCOL COMPLEXITY:** DER Gateways must be configured separately for each protocol
- ❖ **UTILITIES OVERHEAD :** Utilities face increased operational burden - configuration, maintenance, troubleshooting whenever DER scales.

PROPOSED SOLUTION FOR PROTOCOL TRASLATION

- ❖ **DER Protocol Bridge:** A hybrid Python/Rust module for real-time, bidirectional translation for the Local DER Communication Interface (LDERCI).
- ❖ **JSON-based Mapping:** Encapsulates DER data in standardized JSON payloads, mapping SunSpec Modbus registers ↔ DNP3 (IEEE 1815) data.
- ❖ **Deployment on DER Gateway:** Runs as a Linux-based gateway on NVIDIA Jetson Orin, between DER devices and upstream DERMS/utility interfaces.
- ❖ **Performance Evaluation:** Quantify translation latency, throughput, and smart-inverter function conformance (e.g., Volt-Var curve).

METHODOLOGY FOR BUILDING DER NETWORK

Emulate a Python-based **Sunspec Compliant DER Device** using pymodbus[3]

Build a **Sunspec Modbus Client** using pysunspec2

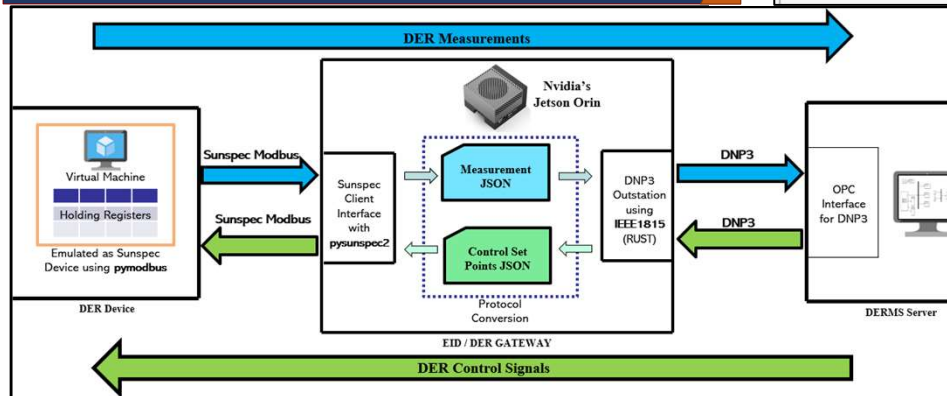
Develop an IEEE 1815 **Rust-based DNP3 outstation** interface

Design and develop the DER-PB to **parse and map DER data** using JSON[4]

Integrate with **MATLAB DERMS** to emulate Utility Control

To conduct **Performance and Conformance Testing**

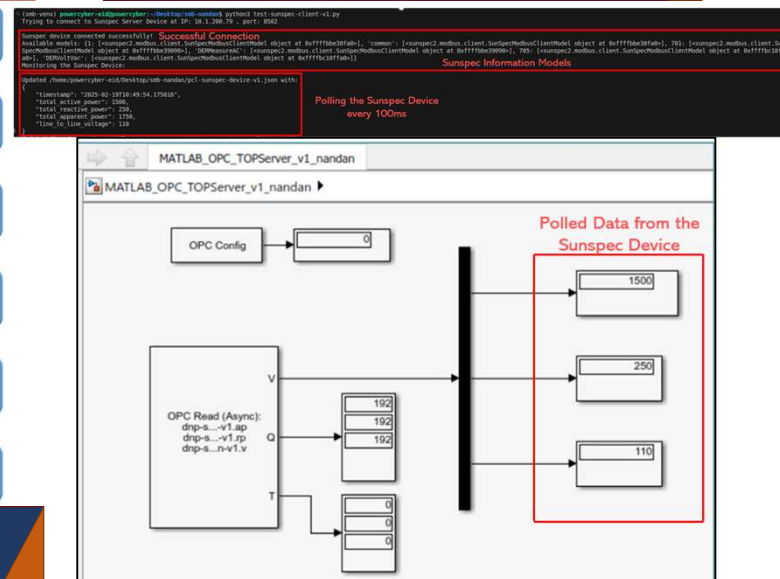
EXPERIMENTAL EVALUATION OF IN THE DER NETWORK



CONCLUSION

- ❖ **Interoperable DER Network:** A Real-time, bidirectional conversion between Sunspec Modbus & IEEE 1815 (DNP3).
- ❖ **Smart Inverter Conformance:** Validated limit active P, Constant Q, Constant PF, V-VAR, V-HZ.
- ❖ **Edge-Ready Deployment:** DER PB running on Nvidia Jetson Orin as a lightweight software enabling turnkey integration and future protocol extensions.

RESULTS AND VALIDATION



FUTURE WORK

- ❖ **Secure TLS Tunnel:** Enabling end-to-end TLS 1.3 for all DER Protocol Interfaces to ensure CIA Properties
- ❖ **IEEE 2030.5 Extension:** Implementing IEEE 2030.5 Interface in WAN.
- ❖ **Public Access:** All the developed products will be made available through **GIT** for other researchers.

REFERENCES

- [1] [Online]. Available: <https://sunspec.org/wp-content/uploads/2025/01/SunSpec-DER-Information-Model-Specification-V1-2-1.pdf>
- [2] [Online]. Available: <https://github.com/pymodbus-dev/pymodbus>
- [3] [Online]. Available: <https://github.com/sunspec/pysunspec2>
- [4] [Online]. Available: <https://github.com/stepfunc/dnp3>
- [5] [Private] Available: <https://github.com/hounandan-isu/sunspec-modbus-server-v1.git>

